

JMO – 2008

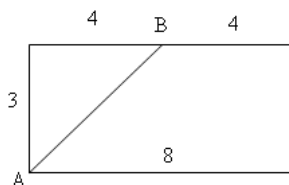
Max Mark – 96

Time – 3 Hrs

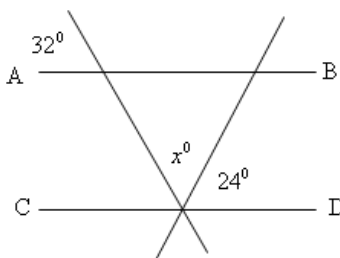
Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- All questions are of equal value.
- Answer all the questions.

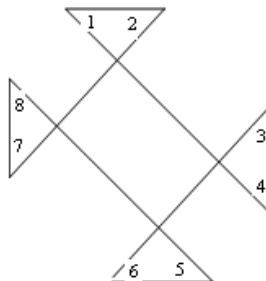
1.
 - (a) A shop advertises: Buy one, get another at half price. What is the net discount in percent?
 - (b) If the square of a non zero number is 70% of the number. What is original number?
2.
 - (a) If $1 - 2 + 3 - 4 + 5 - 6 + \dots + (n - 2) - (n - 1) + n = 2008$, then what is the value of n ?
 - (b) Find the value of $(1 + \frac{1}{2})(1 + \frac{1}{3})(1 + \frac{1}{4}) \dots (1 + \frac{1}{2008})$.
3.
 - (a) If two sides of a right angled triangle are 5 cm and 6 cm long, how many possibilities are there for the third side?
 - (b) How many boxes are crossed by a diagonal of a rectangular table formed by 199X991 small squares?
4.
 - (a) Cross out 10 digits from the number 1234512345123451234512345 so that remaining number is as large as possible.
 - (b) What is the digit in the unit place of the number 2009^{2009} ?
5. A 3X8 rectangle is cut into two pieces along the line AB as shown in the figure. The two pieces are then rearranged to form a right angled triangle. What is the perimeter of the triangle formed?



6. A seven digit number is such that each group of four adjacent digits add to 16 and each group of five adjacent digits add to 19. What is the sum of the digits of the number?
7. Find the smallest natural number which yields remainders 1, 2, 3, 4 & 5 when divided by 2, 3, 4, 5, & 6 respectively.
8. What is the acute angle between the minute and hour hand of clock at 2.30?
9.
 - (a) In the figure below AB and CD are parallel lines. Find value of x .



(b) In the figure below what is the sum of angles marked 1, 2, 3, 4, 5, 6, 7 and 8.



- 10.
- (a) A's monthly income is 20% more than B's monthly income. Find how much percent B's monthly income is less than that of A?
 - (b) If Sam gave Ram Rs. 5 then Ram would have 5 times as much as Sam, but if Ram gave Sam Rs. 5 then Sam would have 5 times as much as Ram. How much did they have together?
11. An article was sold at a profit of 12%. If the article would have purchased 10% less and sold at Rs. 6 more then the profit would have 25%. What is the purchase cost of the article?
 12. If a , b , c and d represent distinct non zero digits for which $aaaa+bbb+cc+d=1995$, then find $a \times b \times c \times d$.
 13. If Ram goes from his home to school at a speed of 2 km/h then he is 6 minute late, but if he goes at a speed of 3 km/h then he reaches school 10 min early. What is the distance between his school from his home?
 14. Out of 5 questions in a paper 5% of the candidates answered all and 5% answered none. Of the rest 25% answered only one question and 20% answered four questions. If $24\frac{1}{2}\%$ of the total candidates answered only two questions and 200 candidates answered only 3 questions, find out total number of candidates.
 15. A man started from home at 2.35PM (by his clock at home) and drove to his village and arrived there when the village clock indicated 3.10PM. After staying there for 25 minutes he drove at 4 PM (by his clock at home). Find out how much the village clock was fast or late in comparison to his clock at home?
 16. Out of 27 similar looking coins, one is fake coin whose weight is less than that of a genuine coin. If a beam balance is given to you (without any weight) chalk out a strategy to detect the fake coin by using balance exactly thrice.

**